Table 2. Guinea-Bissau: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2018–38 (concluded)

(In percent)

	Projections							
	2018	2019	2020	2021	2022	2023	**2028**	2038
Debt service-to-exports ratio								
Baseline	2	2	2	2	2	2	**2**	3
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	2	2	2	1	1	2	**1**	-1
A2. New public sector loans on less favorable terms in 2018-2038 2	2	2	2	2	2	2	**3**	5
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	2	2	2	2	2	2	**2**	3
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	2	2	4	4	4	5	**5**	9
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	2	2	2	2	2	2	**2**	3
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	2	2	2	2	2	2	**2**	3
B5. Combination of B1-B4 using one-half standard deviation shocks	2	2	2	2	2	3	**2**	5
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	2	2	2	2	2	2	**2**	3
Debt service-to-revenue ratio								
Baseline	4	3	3	2	2	3	**2**	2
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2018-2038 1/	4	3	3	2	2	2	**1**	-1
A2. New public sector loans on less favorable terms in 2018-2038 2	4	3	3	3	3	3	**3**	3
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2019-2020	4	3	3	2	2	3	**2**	2
B2. Export value growth at historical average minus one standard deviation in 2019-2020 3/	4	3	3	3	3	3	**2**	3
B3. US dollar GDP deflator at historical average minus one standard deviation in 2019-2020	4	3	3	3	3	3	**2**	2
B4. Net non-debt creating flows at historical average minus one standard deviation in 2019-2020 4/	4	3	3	3	3	3	**2**	2
B5. Combination of B1-B4 using one-half standard deviation shocks	4	3	3	3	3	3	**2**	3
B6. One-time 30 percent nominal depreciation relative to the baseline in 2019 5/	4	4	4	3	3	4	**3**	3
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	47	47	47	47	47	47	**47**	47

Sources: Country authorities; and staff estimates and projections.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly a an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

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Table 3. Guinea-Bissau: Public Sector Sustainability Framework, Baseline Scenario, 2015–38

(In percent of GDP unless otherwise indicated)

	Actual					Estimate	Projections								
	2015	2016	2017	Average 5/	Standard Deviation 5/	2018	2019	2020	2021	2022	2023	2018-23 Average	2028	2038	2024-38 Average
Public sector debt 1/	52.4	53.9	50.8			50.9	49.4	47.9	46.5	45.1	43.7		40.1	35.5	
of which: foreign-currency denominated	15.1	14.7	12.4			12.0	12.2	12.8	14.1	15.3	16.1		23.1	11.5	
Net public debt	50.8	52.8	49.1			49.0	47.7	46.3	45.0	43.7	42.4		39.3	35.1	
Change in public sector debt	-2.6	1.5	-3.1			0.1	-1.5	-1.5	-1.4	-1.4	-1.4		-0.5	-0.2	
Identified debt-creating flows	-2.7	-1.2	-8.8			-0.4	-1.1	-1.3	-1.2	-1.2	-1.3		-0.5	-0.2	
Primary deficit	2.9	3.5	1.0	2.5	1.3	2.2	1.9	1.7	1.8	1.8	1.6	1.8	2.1	2.2	2.0
Revenue and grants	20.2	16.0	18.3			18.8	20.2	21.0	21.7	22.5	23.3		29.8	43.9	
of which: grants	6.5	4.0	5.5			4.7	5.0	5.1	5.2	5.3	5.4		5.4	5.4	
Primary (noninterest) expenditure	23.1	19.5	19.3			21.0	22.1	22.8	23.5	24.2	25.0		31.9	46.2	34.8
Automatic debt dynamics	-6.3	-4.8	-6.8			-2.5	-3.0	-3.0	-3.0	-3.0	-2.9		-2.6	-2.5	
Contribution from interest rate/growth differential	-6.7	-4.6	-4.9			-2.2	-2.9	-2.9	-2.9	-2.9	-2.8		-2.4	-2.4	
of which: contribution from average real interest rate	-3.5	-1.5	-1.9			0.4	-0.4	-0.5	-0.6	-0.6	-0.6		-0.5	-0.7	
of which: contribution from real GDP growth	-3.2	-3.1	-3.0			-2.6	-2.5	-2.4	-2.3	-2.2	-2.1		-1.9	-1.7	
Contribution from real exchange rate depreciation	0.3	-0.2	-1.9			-0.4	-0.1	-0.1	-0.2	-0.2	-0.2		...	...	
Other identified debt-creating flows	0.7	0.0	-3.0			-0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Privatization receipts (negative)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Recognition of implicit or contingent liabilities	0.7	0.0	0.2			-0.1	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Debt relief (HIPC and other)	0.0	0.0	-3.2			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Other (specify, e.g. bank recapitalization)	0.0	0.0	0.0			0.0	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
Residual, including asset changes	0.1	2.7	5.7			0.5	-0.4	-0.2	-0.2	-0.2	-0.1		0.0	0.0	
Other Sustainability Indicators															
PV of public sector debt	...	...	47.5			47.7	46.3	44.4	42.2	40.2	38.3		31.8	32.4	
of which: foreign-currency denominated	...	...	9.0			8.8	9.1	9.3	9.8	10.4	10.8		14.7	8.3	
of which: external	...	...	9.0			8.8	9.1	9.3	9.8	10.4	10.8		14.7	8.3	
PV of contingent liabilities (not included in public sector debt)	...	...	...			...	...	...	...	...	...		...	...	
Gross financing need 2/	11.1	10.1	9.8			7.6	6.7	5.7	5.4	4.9	4.4		4.1	3.6	
PV of public sector debt-to-revenue and grants ratio (in percent)	...	...	259.7			253.2	229.6	210.8	194.5	179.1	164.2		106.6	73.8	
PV of public sector debt-to-revenue ratio (in percent)	...	...	370.5			337.9	304.4	278.4	255.8	234.5	213.8		130.3	84.1	
of which: external 3/	...	...	70.4			62.5	59.6	58.1	59.6	60.6	60.1		60.4	21.7	
Debt service-to-revenue and grants ratio (in percent) 4/	6.6	8.0	24.0			7.8	7.1	5.8	5.7	5.2	4.9		3.2	2.1	
Debt service-to-revenue ratio (in percent) 4/	9.8	10.7	34.3			10.4	9.4	7.6	7.5	6.9	6.4		3.9	2.4	
Primary deficit that stabilizes the debt-to-GDP ratio	5.5	2.0	4.1			2.1	3.4	3.3	3.2	3.2	3.0		2.6	2.5	
Key macroeconomic and fiscal assumptions															
Real GDP growth (in percent)	6.1	6.3	5.9	4.0	2.9	5.3	5.1	5.0	5.0	5.0	5.0	5.1	5.0	5.0	5.0
Average nominal interest rate on forex debt (in percent)	1.0	2.2	1.3	0.1	1.3	1.3	1.2	1.7	2.1	2.2	2.4	1.8	2.5	3.2	2.8
Average real interest rate on domestic debt (in percent)	-9.2	-4.5	-4.6	-2.7	5.4	...	-0.7	-1.4	-1.7	-2.2	-2.4	-1.7	-3.4	-3.7	-3.5
Real exchange rate depreciation (in percent, + indicates depreciation	2.3	-1.2	-14.0	-0.7	8.0	-3.1	...	...	...	...	...	...	...	...	...
Inflation rate (GDP deflator, in percent)	12.1	6.1	5.9	4.9	6.2	0.1	2.2	2.4	2.5	2.8	2.8	2.1	2.8	2.8	2.8
Growth of real primary spending (deflated by GDP deflator, in percer	-0.7	-10.0	4.5	-0.5	3.7	14.9	10.4	8.3	8.4	8.3	8.0	9.7	10.4	11.8	9.4
Grant element of new external borrowing (in percent)	...	...	...	...	...	41.1	46.7	49.6	49.6	49.6	49.6	47.7	46.9	45.8	...

Sources: Country authorities; and staff estimates and projections.

1/ Comprises public and publicly guaranteed central government debt on a gross basis.
2/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period.
3/ Revenues excluding grants.
4/ Debt service is defined as the sum of interest and amortization of medium and long-term debt.
5/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

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Table 4. Guinea-Bissau: Sensitivity Analysis for Key Indicators of Public Debt, 2018–38

	Projections							
	2018	2019	2020	2021	2022	2023	2028	2038
PV of Debt-to-GDP Ratio								
Baseline	48	46	44	42	40	38	32	32
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	48	47	45	43	41	40	33	34
A2. Primary balance is unchanged from 2018	48	46	45	43	41	40	33	34
A3. Permanently lower GDP growth 1/	48	47	45	43	42	40	37	52
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	48	48	48	46	45	43	40	47
B2. Primary balance is at historical average minus one standard deviations in 2019-202	48	47	47	44	42	40	33	33
B3. Combination of B1-B2 using one half standard deviation shocks	48	48	47	45	43	42	37	41
B4. One-time 30 percent real depreciation in 2019	48	50	47	45	42	40	31	33
B5. 10 percent of GDP increase in other debt-creating flows in 2019	48	54	51	48	46	43	36	35
PV of Debt-to-Revenue Ratio 2/								
Baseline	253	230	211	195	179	164	107	74
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	253	231	214	198	184	170	111	77
A2. Primary balance is unchanged from 2018	253	231	213	198	184	170	111	77
A3. Permanently lower GDP growth 1/	253	231	213	198	184	171	122	117
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	253	236	225	210	197	183	131	107
B2. Primary balance is at historical average minus one standard deviations in 2019-202	253	236	223	205	188	172	111	76
B3. Combination of B1-B2 using one half standard deviation shocks	253	236	222	207	192	178	122	94
B4. One-time 30 percent real depreciation in 2019	253	246	225	205	187	170	104	76
B5. 10 percent of GDP increase in other debt-creating flows in 2019	253	268	244	223	204	186	120	79
Debt Service-to-Revenue Ratio 2/								
Baseline	8	7	6	6	5	5	3	2
A. Alternative scenarios								
A1. Real GDP growth and primary balance are at historical averages	8	7	6	6	6	5	3	2
A2. Primary balance is unchanged from 2018	8	7	6	6	6	6	4	2
A3. Permanently lower GDP growth 1/	8	7	6	6	6	5	5	6
B. Bound tests								
B1. Real GDP growth is at historical average minus one standard deviations in 2019-20	8	7	6	7	7	7	5	5
B2. Primary balance is at historical average minus one standard deviations in 2019-202	8	7	6	7	7	6	3	2
B3. Combination of B1-B2 using one half standard deviation shocks	8	7	6	7	7	6	5	4
B4. One-time 30 percent real depreciation in 2019	8	8	7	7	7	6	5	4
B5. 10 percent of GDP increase in other debt-creating flows in 2019	8	7	8	15	9	8	4	3

Sources: Country authorities; and staff estimates and projections.
1/ Assumes that real GDP growth is at baseline minus one standard deviation divided by the square root of the length of the projection period.
2/ Revenues are defined inclusive of grants.

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